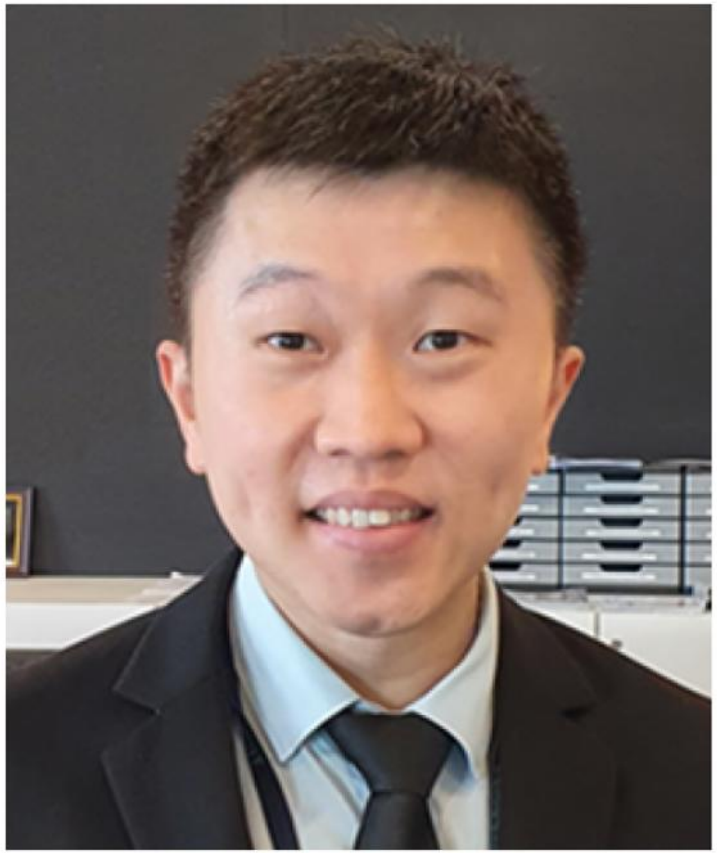
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Renewable Generation Technologies (EEEE3045)

Introduction



Yousif Abdalla Abakar (PhD)
Professor of Renewable Energy



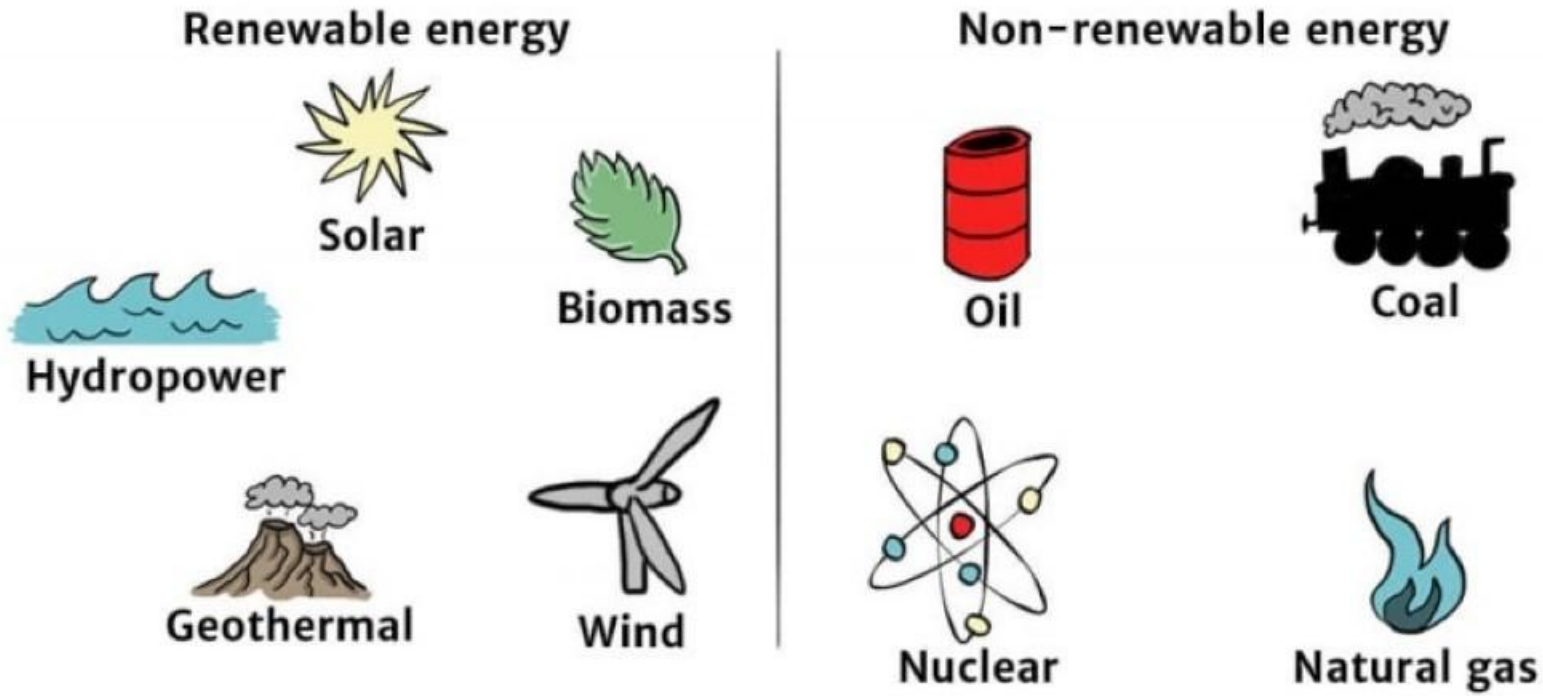


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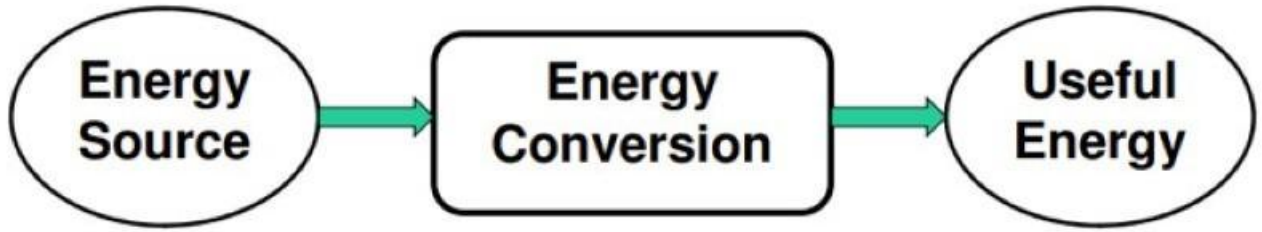
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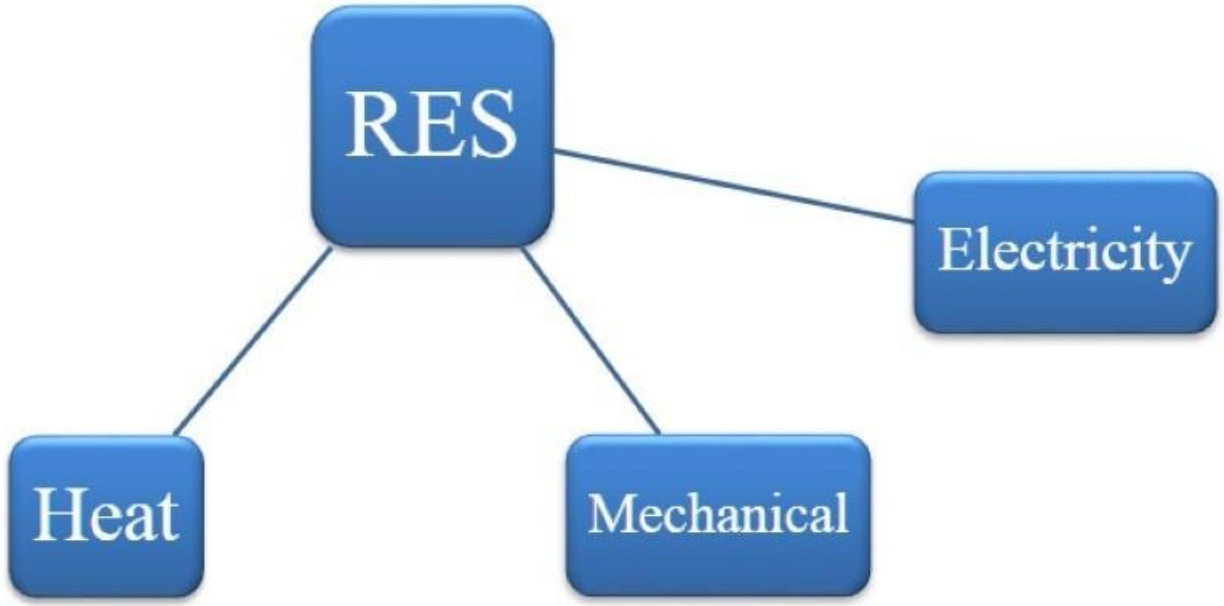
Renewable Energy

- What is Renewable Energy?
- Why Renewable Energy is important?
- What are the main sources of Renewable Energy?



What does Energy Conversion mean?





Historic Energy Conversion Sequences

- Biomass → heat (esp. cooking)
- Solar → heat, dry clothes, dry food
  - Solar is still main light source, no need for conversion
  - Solar is source of biomass, wind, hydro, etc.
- Biomass → farm animals → horsepower, food

Later, people also did these conversions:

- Coal → heat
- Hydro → milling flour, running machinery
- Wind → pump water

Modern Energy Conversion Sequences

Heating of Buildings:

- Gas, oil, biomass → heat
- Solar → heat

Electricity Generation:

- Coal, gas, nuclear → heat → mechanical → electricity
- Hydro → mechanical → electricity
- Wind → mechanical → electricity
- Solar → Electricity

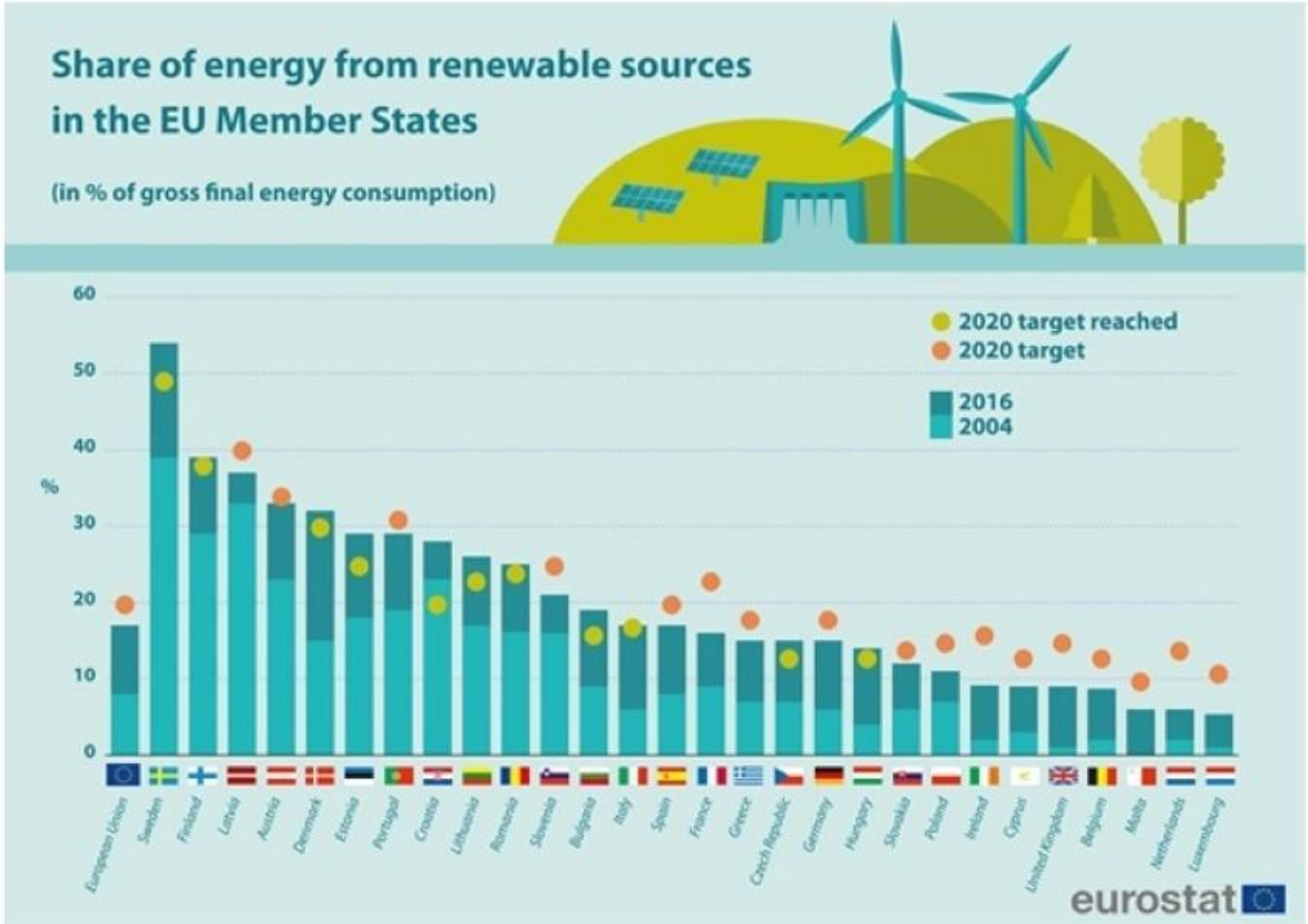
Transportation:

- Oil → gasoline, diesel, jet fuel → heat → mechanical
- Biomass → ethanol → heat → mechanical
- Fuel cell cars: Gas → hydrogen → electricity → mechanical
- Hybrid cars: Gasoline → mechanical → electricity → battery → electricity → mechanical

Political Perspective

- Renewable energy resources can be divided into numerous categories depending upon the political goals or objectives under consideration.

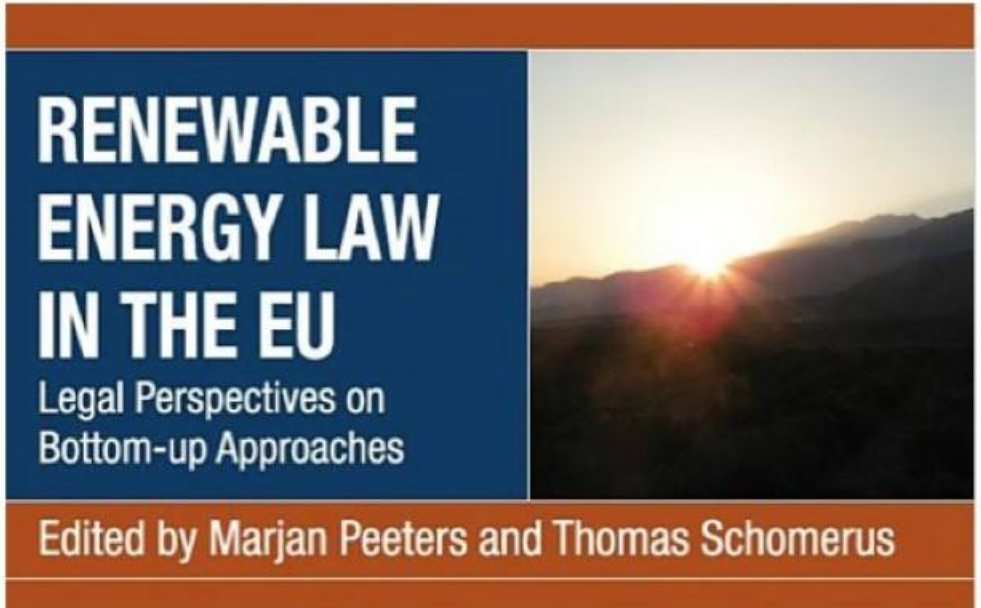
What has politics got to do with Renewable Energy?
Do Governments deliver what the public want?
Can Renewable Energy thrive without political support?
YOU CAN FIND MANY ARTICLES ABOUT THIS TOPIC



Legal perspective

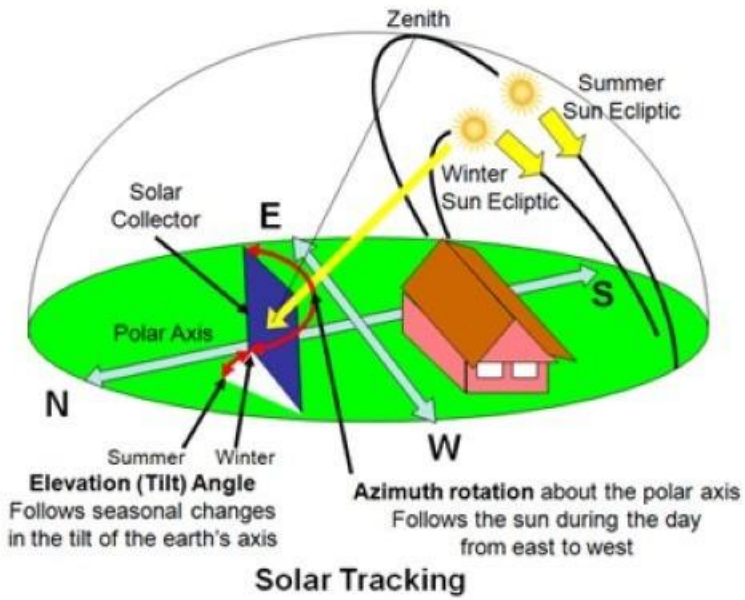
- Existing laws such as land use, water, mining, and hydrocarbon laws need to be scrutinized to determine their potential jurisdiction over and applicability to renewable resources.
- It is important to define what technologies are to be considered "renewable" for the purposes of any piece of legislation.

Renewable Energy Law in the EU: Legal Perspectives on Bottom-Up Approaches (New Horizons in Environmental and Energy Law series)



Main Topics of RGT Module

- Sun radiation and different kinds of electricity generation using solar energy.
- PV (PV system components)
- Designing grid-connected and off-grid PV systems





Main Topics of RGT Module

- What is Wind Energy?
- Wind Energy potential in the world
- Basic equations
- Wind turbines
- Wind farms



Main Topics of RGT Module

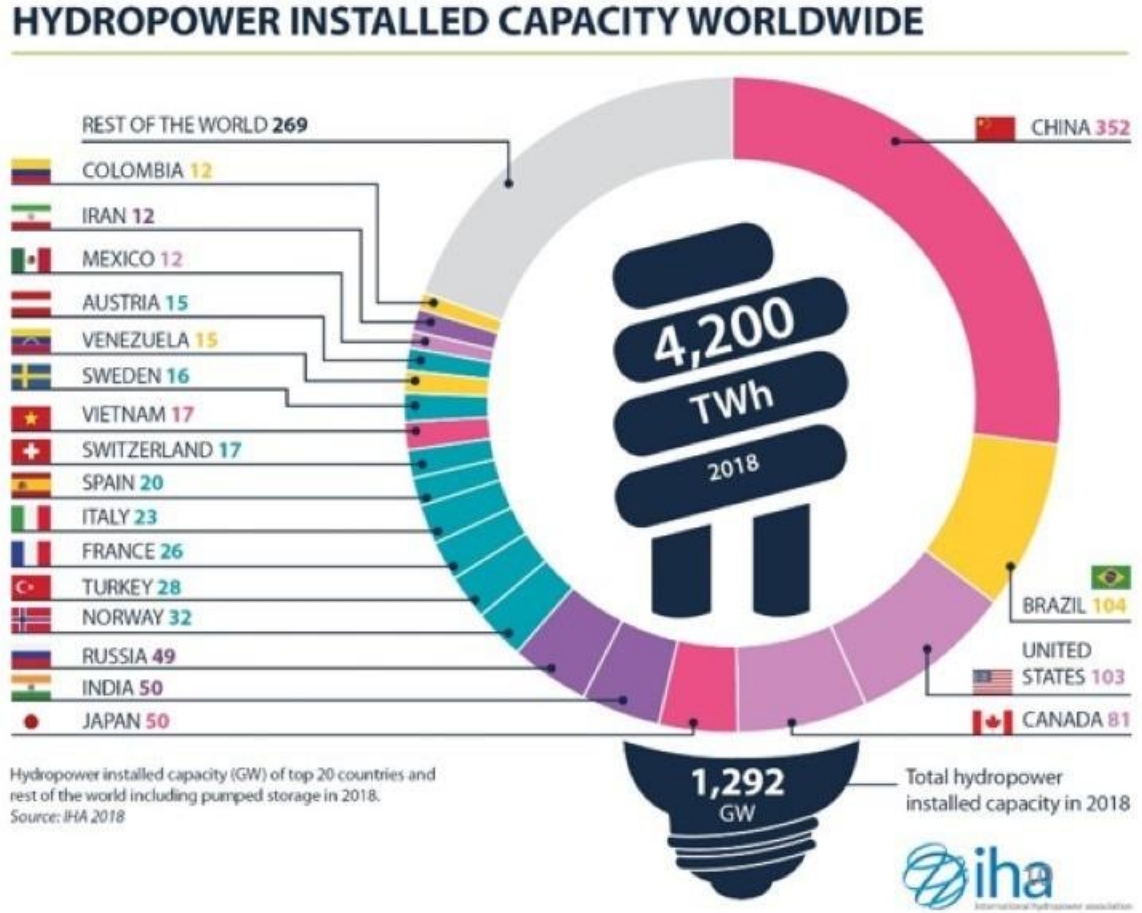
- Hydro Power & Variable Speed Pumped Hydro
- Types of Hydro power plant
- Different turbine types





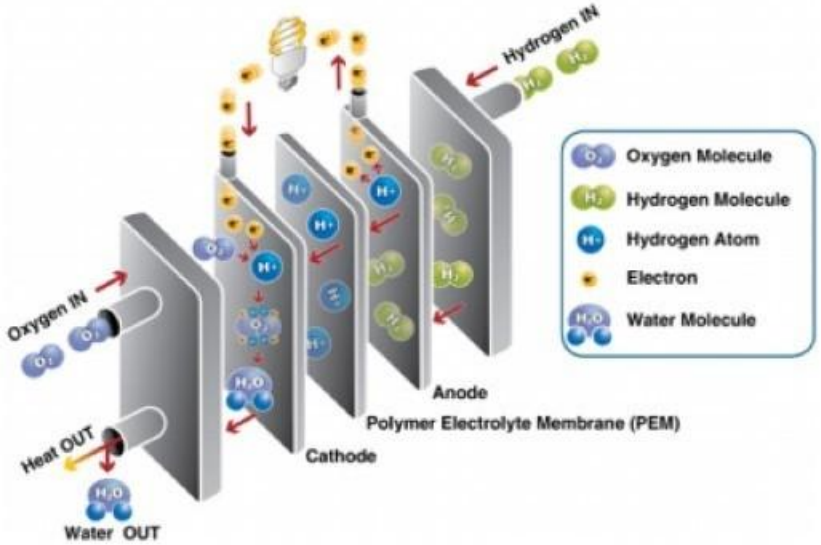
Fig 6: Pelton Wheel





Main Topics of RGT Module

- Biomass
- Waste
- Geothermal
- Fuel Cell
- Generators

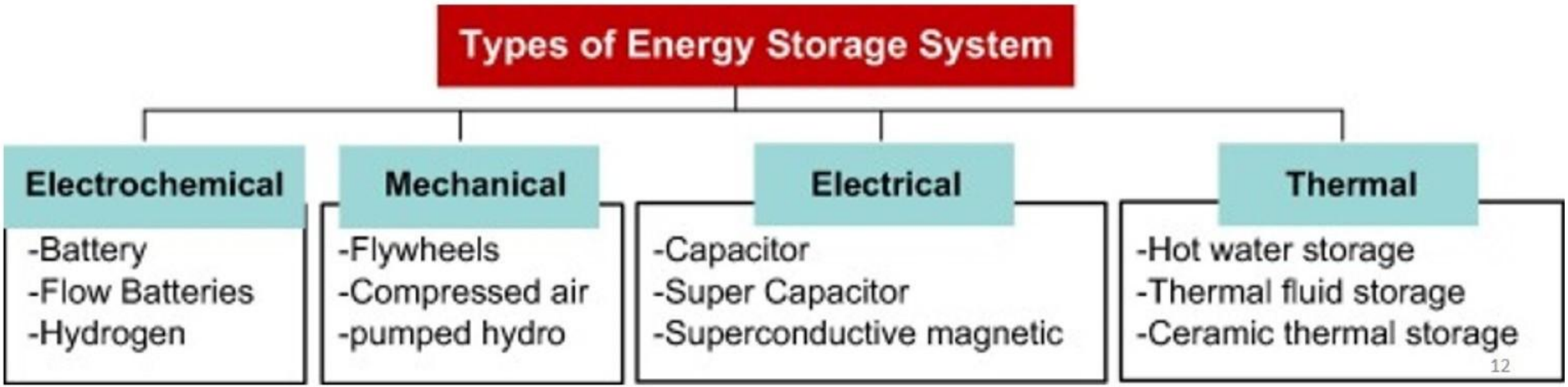






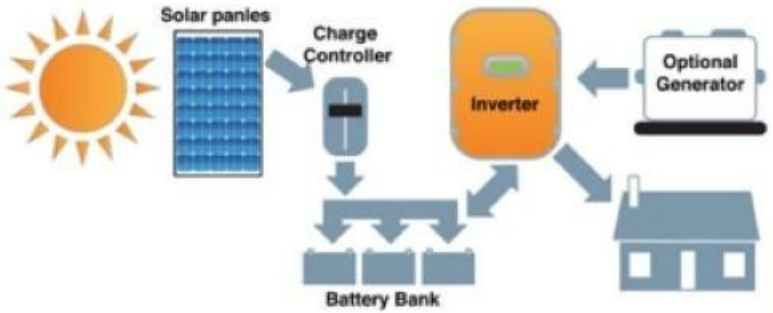
Main Topics of RGT Module

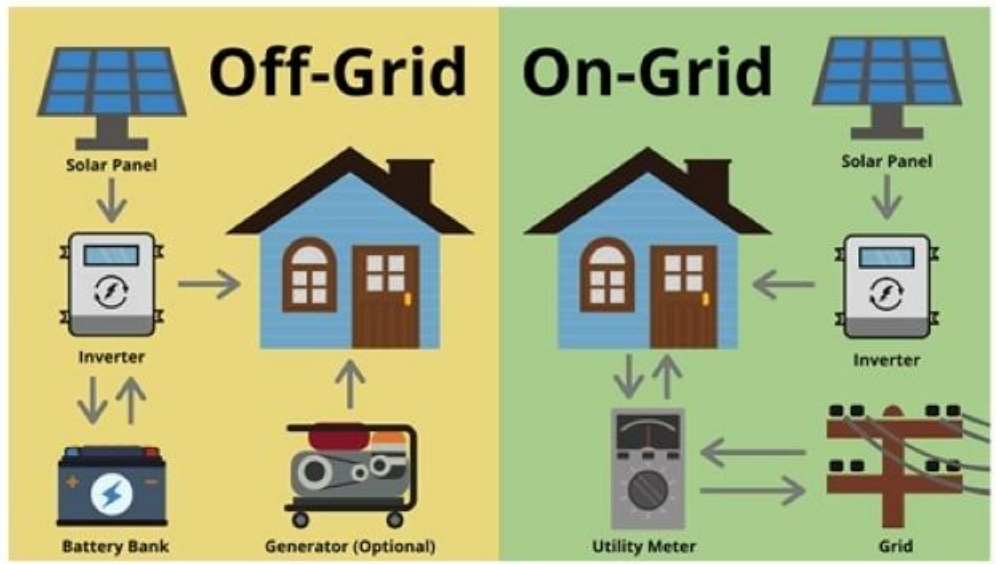
- Energy Storage
- RE in the Grid (+power quality)
- Renewable Generation & Control



What are the renewable energy applications?

- Renewable energy applications generally break down into two categories or applications, “on-grid” and “off-grid”.







Does renewable energy generate affordable power?

- Universally, the goal of electric power generation planners is to deliver electricity to the maximum number of customers at the lowest possible price.
- The political acceptability of power generated from any source will depend upon the ultimate tariff to the consumer relative to the benefits delivered.

Learning Outcomes

1. Evaluate different type of energy sources and understand their trends, which may include conventional and non-conventional options depending on the context and application.
2. Select appropriate type of renewable energy source and energy storage to mitigate risks associated with security of electrical supply for a given application
3. Analyse various energy storage systems associated with renewable power generation.
4. Determine high level system sizing and component understanding for the renewable power generation.
5. Demonstrate an understand various renewable energy sources to design and analyse an integrated renewable energy system.
6. Analyse options (i.e. policies, advices or suggestions) aimed at improving efficiency in renewable energy generation and consumption.
7. Develop pragmatic knowledge of renewable energy sources and systems.

Method and Frequency of Class

Activity	Number of weeks	Number of sessions per week	Duration of a session
Lecture	12 weeks	1	2 hours
Tutorial	8 weeks	1	1 hour
Laboratory	2 weeks (1 group)	1	3 hours

Method of Assessment

Assessment Type	Weight (%)
Final Exam	50
Coursework 1 (HOMER Report)	30
Coursework 2 (Lab Reports)	20

End

